

Algebra — revision of Test 1

Lessons 1-8 · Quarter I · 60 questions

Work down each column. Where a fraction is cancelled, the permissible values are part of the answer — an answer without them is half an answer.

Lessons 1-3 · Revision of the Grade 7 course 22

Indices, brackets, factorising

- 1

$x^7 \cdot x^5$
 x^{12}
Same base — add the indices.
- 2

$a^9 \div a^4$
 a^5
Same base — subtract the indices.
- 3

$(b^4)^3$
 b^{12}
Power of a power — multiply the indices.
- 4

$(2x^3)^4$
 $16x^{12}$
Raise the 2 as well: $2^4 = 16$.
- 5

$(-3a^2b)^2$
 $9a^4b^2$
A negative squared is positive.
- 6

$5x^2 \cdot 4x^5$
 $20x^7$
- 7

$\frac{12a^6}{3a^2}$
 $4a^4$
- 8

Expand $3(2x - 7)$
 $6x - 21$
- 9

Expand $-5(a - 4)$
 $-5a + 20$
The minus changes both signs.
- 10

Expand $x(x + 6)$
 $x^2 + 6x$

- 11

Expand $2a(3a - 5)$
 $6a^2 - 10a$
- 12

Expand $(x + 3)(x + 8)$
 $x^2 + 11x + 24$
- 13

Expand $(y - 5)(y + 2)$
 $y^2 - 3y - 10$
- 14

Expand $(2x - 1)(x + 4)$
 $2x^2 + 7x - 4$
- 15

Expand $(a + 7)^2$
 $a^2 + 14a + 49$
Do not forget the middle term.
- 16

Expand $(3m - 2)^2$
 $9m^2 - 12m + 4$
- 17

Expand $(x - 6)(x + 6)$
 $x^2 - 36$
Difference of two squares.
- 18

Factorise $8x + 20$
 $4(2x + 5)$
- 19

Factorise $6a^2 - 9a$
 $3a(2a - 3)$
Take out $3a$, not just 3 .
- 20

Factorise $x^2 - 49$
 $(x - 7)(x + 7)$
- 21

Factorise $25m^2 - 4n^2$
 $(5m - 2n)(5m + 2n)$
- 22

Factorise $x^2 + 10x + 25$
 $(x + 5)^2$

Lessons 4-5 · Algebraic expressions 16

Powers of products, factorising, simplifying

- 23

$(2a^2b^3)^3$
 $8a^6b^9$
- 24

$(-x^4y)^3$
 $-x^{12}y^3$
An odd power keeps the minus.
- 25

$(5p^3)^2 \cdot 2p$
 $50p^7$
- 26

$\frac{a^5b^2}{a^2b}$
 a^3b
- 27

Factorise $3x^2 - 12$
 $3(x - 2)(x + 2)$
Common factor first, then the squares.
- 28

Factorise $2a^2 - 18b^2$
 $2(a - 3b)(a + 3b)$
- 29

Factorise $x^3 - x$
 $x(x - 1)(x + 1)$
- 30

Factorise $4y^2 - 20y + 25$
 $(2y - 5)^2$

- 31

Factorise $ax + ay + bx + by$
 $(a + b)(x + y)$
Group in pairs.
- 32

Factorise $m^2 - m - 12$
 $(m - 4)(m + 3)$
- 33

Factorise $x^2 + 7x + 12$
 $(x + 3)(x + 4)$
- 34

Factorise $5x^2y - 10xy^2$
 $5xy(x - 2y)$
- 35

Simplify $3(x - 2) - 2(x - 5)$
 $x + 4$
- 36

Simplify $(a + 4)^2 - (a - 4)^2$
 $16a$
- 37

Simplify $(x + 2)(x - 2) + 4$
 x^2
- 38

Find $x^2 - 4x$ when $x = -2$
 12
 $4 + 8$. Watch the sign.

39 Permissible values of $\frac{1}{x-9}$
 $x \neq 9$

40 Permissible values of $\frac{5}{2x+6}$
 $x \neq -3$

41 Permissible values of $\frac{x+1}{x^2-4}$
 $x \neq 2, x \neq -2$

42 Permissible values of $\frac{3}{x^2+1}$
every x
 x^2+1 is never zero, so nothing is excluded.

43 Permissible values of $\frac{x-2}{x(x-7)}$
 $x \neq 0, x \neq 7$

44 Find $\frac{x+7}{x-3}$ when $x=5$
6

45 Find $\frac{2x-1}{x+4}$ when $x=-1$
-1

46 Find $\frac{x^2-9}{x+3}$ when $x=4$
1

47 For which x is $\frac{x-8}{x+2}$ zero?
 $x=8$
Numerator zero, denominator not.

48 For which x is $\frac{3x+6}{x-1}$ zero?
 $x=-2$

49 Cancel $\frac{24x^5}{6x^2}$
 $4x^3, x \neq 0$

50 Cancel $\frac{14a^3b}{7ab}$
 $2a^2, a \neq 0, b \neq 0$

51 Cancel $\frac{5x+15}{x+3}$
5, $x \neq -3$

52 Cancel $\frac{x^2-16}{x-4}$
 $x+4, x \neq 4$

53 Cancel $\frac{x^2-25}{x^2+5x}$
 $\frac{x-5}{x}, x \neq 0, x \neq -5$

54 Cancel $\frac{2x+8}{x^2-16}$
 $\frac{2}{x-4}, x \neq 4, x \neq -4$

55 Cancel $\frac{x^2+6x+9}{x^2-9}$
 $\frac{x+3}{x-3}, x \neq 3, x \neq -3$

56 Cancel $\frac{3a-3b}{b-a}$
-3, $a \neq b$
 $b-a=-(a-b)$ — that swap is where the minus comes from.

57 Cancel $\frac{x^2-x}{1-x}$
 $-x, x \neq 1$

58 Cancel $\frac{4x^2-9}{2x+3}$
 $2x-3, x \neq -\frac{3}{2}$

59 Cancel $\frac{x^3-4x}{x^2+2x}$
 $x-2, x \neq 0, x \neq -2$

60 Why can $\frac{x+3}{3}$ not be cancelled?
the 3 is a term, not a factor
Only a factor of the *whole* numerator may cancel.